Yulei Fu

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EDUCATION

University of Michigan Sep 2023 - May 2027
Bachelors of Science in Robotics Engineering, GPA 3.6 Ann Arbor, MI

SKILLS

Languages: C, C++, Rust, Python, Bash, Verilog
Software Tools: Embedded Linux, Yocto, FreeRTOS, D-Bus, IPC, MQTT, GDB, Docker, Google Test, CMake
Robotics: ROS2, Nav2, Google Cartographer, Slam Toolbox, Cyclone DDS
Relevant Coursework: Robot Operating Systems (ROS2), Logic Design, Data Structures & Algorithms, Embedded Systems

EXPERIENCE

H3D, Inc. June 2026 - August 2026
Robotics Engineering Intern Ann Arbor, MI

- Served as sole engineer on a federal SBIR Phase I program, delivering an autonomous radiation-mapping UGV for nuclear plant surveys; work secured a Phase II award exceeding $1M as of July 2026.
- Built and integrated full sensor stack on an NVIDIA Jetson AGX Orin (ConnectTech Rogue carrier, dual 10GbE), fusing Ouster OS0-128 LiDAR over dedicated ethernet, RealSense D455 depth camera, WHEELTEC STM32F103RC differential-drive base, and H3D CZT gamma-ray spectrometer for real-time autonomous radiation mapping.
- Designed a 3D voxel mapping layer post-processing timestamped gamma-ray measurements with Cartographer poses to generate spatial dose-rate heatmaps for contamination localization.
- Integrated Google Cartographer SLAM into a 23-package ROS2 workspace, configuring Ceres-based non-linear scan matching and hierarchical submaps while building and debugging its 7-dependency C++ stack from source.

Boeing Jan 2026 - June 2026
Simulation Intern Hybrid

- Built IsaacSim digital twin models in Docker to virtually commission 2 manufacturing configurations, eliminating 6-10 physical test iterations and cutting per-configuration evaluation time by 85% (168 hrs to 15 hrs).
- Evaluated 3 robotic agent configurations (HSR, KMR, RS007) in Docker for virtual commissioning across 2 manufacturing layouts using simulated cycle-time and workspace constraints to identify feasible human-robot process reconfigurations.
- Developed a CAD-to-URDF-to-USD asset conversion and scene composition pipeline for dynamic environment generation in IsaacSim, enabling reproducible reconfiguration across varying part geometries.

Amazon Robotics Jan 2025 - June 2025
Embedded Software Engineer Co-Op Westborough, MA

- Delivered a C++ telemetry and log-capture pipeline for a safety-certified embedded Linux controller, publishing 10K+ messages/day to AWS S3 over MQTT for fleet data persistence in S3; reduced data loss by 94% and issue reproduction time by 85% (6 hrs to 45 min).
- Developed a multithreaded C++ D-Bus service on Yocto Linux to aggregate system state across 3+ subsystems, implementing deterministic IPC and safe message batching for a safety-certified robot controller.
- Shipped a remote Wi-Fi diagnostics API, adding to the system's hardware-diagnostics suite, reducing field-troubleshooting from 2+ hours to 2 minutes for on-sites that previously had no remote pathway to inspect connectivity issues.

Stirling Research Group Nov 2024 - Jan 2025
EVA Research Intern Ann Arbor, MI

- Implemented the Keytel metabolic-rate model in JavaScript on a nRF52840 BLE-based heart-rate wearable to convert real-time heart-rate telemetry into energy-expenditure estimates, enabling accurate British Thermal Unit (BTU) tracking.

PROJECT EXPERIENCE

MBot Planning and Perception (ROS2, C++)

- Developed a ROS2 autonomous exploration stack with a 4-state FSM and clearance-aware A planner, using an obstacle-distance grid and line-of-sight pruning to generate collision-free paths alongside AprilTag and wall-color perception

Passive WiFi Surveillance Detection Firmware (ESP32, C++, FreeRTOS, Python, Flask)

- Designed a lock-free ring buffer between the WiFi sniffer ISR (firing in IRAM on the radio task) and main loop, enabling safe frame handoff without memory allocation or blocking calls in interrupt context on an ESP-32.